

AWS Cloud Migration Financial Model & ROI Guide

CFO-Focused Executive Financial Planning Framework

1. Executive Financial Overview

This document provides a CFO-focused financial modeling framework for AWS cloud migration.

It includes cost comparison models, ROI calculations, payback analysis, and risk-adjusted financial projections.

- 3–5 Year Total Cost of Ownership (TCO) comparison
- Net Present Value (NPV) model
- Internal Rate of Return (IRR) logic
- Payback period calculation
- Sensitivity analysis approach

2. Current State Cost Baseline (On-Premises)

Cost Category	Annual Cost (\$)
Data Center Facilities	500,000
Hardware Depreciation	750,000
Software Licensing	400,000
IT Operations Labor	900,000
Maintenance Contracts	350,000
Total Annual Cost	2,900,000

3. AWS Target State Cost Model (Estimated)

Cost Category	Annual Cost (\$)
Compute (EC2/EKS)	800,000
Storage (S3/EBS)	250,000
Database (RDS)	300,000
Networking	150,000
Support Plan	120,000
Cloud Operations Labor	600,000
Total Annual Cost	2,220,000

4. Annual Cost Savings

Metric	Amount (\$)
On-Prem Annual Cost	2,900,000
AWS Annual Cost	2,220,000
Annual Savings	680,000

5. One-Time Migration Investment

Cost Component	Amount (\$)
Migration Tools & Consulting	1,200,000
Internal Project Team	600,000
Training & Change Management	200,000
Contingency (10%)	200,000
Total Investment	2,200,000

6. ROI & Payback Calculation

ROI Formula: $(\text{Total Net Benefit} - \text{Investment Cost}) / \text{Investment Cost}$

Annual Savings: \$680,000

Total 5-Year Savings: \$3,400,000

Net Benefit after Investment: \$1,200,000

Metric	Value
Investment	\$2,200,000
5-Year Net Benefit	\$1,200,000
ROI	54.5%
Payback Period	3.2 Years

7. Net Present Value (NPV) – 8% Discount Rate

Year	Cash Flow (\$)	Discounted Value (\$)
0	-2,200,000	-2,200,000
1	680,000	629,630
2	680,000	583,000
3	680,000	540,000
4	680,000	500,000
5	680,000	463,000
Total NPV		515,630

8. Strategic Financial Upside (Not Fully Quantified)

- Reduced downtime and revenue loss
- Faster product release cycles
- Improved scalability for growth
- Reduced security breach exposure
- Elimination of hardware refresh cycles

9. Sensitivity Analysis Variables

- Cloud cost variance (+/- 15%)
- Migration cost overrun risk
- Labor reduction assumptions
- Growth in compute consumption
- Discount rate variation (6–10%)

10. Executive Decision Framework

Approve migration if $NPV > 0$, ROI exceeds hurdle rate, and strategic agility benefits align with corporate growth targets.

Financial review should be conducted quarterly during migration and annually post-migration.